Energy from the oceans

One of the many valuable things that the oceans can offer us is energy to fuel our industries, transportation, and modern lives. Vast reserves of oil and gas have been found in the seabed rocks of shallow coastal seas. Oceanic winds can drive turbines that generate electricity, and it may be possible to use the power of tides, currents, and waves in the same way.

WIND POWER

The winds blowing over the sea are stronger and steadier than the winds that blow over land. This makes shallow coastal seas good places to install wind turbines for producing electricity. Some of these offshore wind farms have more than 100 giant turbines, each generating enough power for 150,000 hair dryers.

▸ OFFSHORE WIND FARM
Anchored to the seabed in shallow water, these wind turbines are connected to the shore by undersea electricity cables.

OIL AND GAS

Fossil remains of marine life locked in rocks on the seabed can turn into oil and natural gas—vital fuels and raw materials for industry. They are extracted using drilling rigs such as this one, which either stand on shallow seabeds or float in much deeper water. Modern oil and gas rigs can work in water depths of 9,800 ft (3,000 m), and drill up to 16,400 ft (5,000 m) below the seabed.